

controlled rate to the patient

4. Upon sensing that the lung is filled to the right capacity, the air flow is stopped

5. Thereafter exhale phase is sensed for the patient

6. Once exhale phase is sensed, another air channel is selected for exhaling the air from the lungs to the atmosphere

This is an extremely time-critical operation as inhale and exhale happen within a time span of 1-2 seconds. Even slight delay in computing can hamper the entire process. Hence there is hardly any setting required for the ventilator and the entire process can be managed by the ventilator itself without requiring the services of any health practitioner.

Acceptance and validation

The new ventilator has brought hope to thousands of those patients who have complete body paralysis and



Mechanism inside the ventilator

have been hospitalised for many years. Now, they can afford a ventilator and get back to their homes. This ventilator can be used in all the situations where normal ventilators are used, such as in hospitals, ambulances and even homes.

Prof. Vaish explained, "Although the innovation is breakthrough, we still face some basic functional challenges. Like, in manufacturing, the process to obtain certifications is slow and tedious. We are applying for

major global certifications and hope to get them in the near future. We already have two patents credited for our product."

"Technically, mistakes and innovations go hand-in-hand. We tried various sensors and different algorithms. At times, the members of team thought it was impossible. However, it is through consistent efforts and hard work that we won.

We finally succeeded after over 50 iterations," Dr Vaish added.

The road ahead

The team wants its product to penetrate as deep as possible. It is looking for support from the government as well. It is also open to partnership with other companies, especially those in medical domain. The team hopes that the product reaches to people in need not just in India but throughout the world. It wants the sales to cross 50,000 units next year. **EFY**



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